accepted that some animals are called male, and others female, without asking what these words really meant. But what is the essence of maleness? What, at bottom, defines a female? We as mammals see the sexes defined by whole syndromes of characteristics—possession of a penis, bearing of the young, suckling by means of special milk glands, certain chromosomal features, and so on. These criteria for judging the sex of an individual are all very well for mammals but, for animals and plants generally, they are no more reliable than is the tendency to wear trousers as a criterion for judging human sex. In frogs, for instance, neither sex has a penis. Perhaps, then, the words male and female have no general meaning. They are, after all, only words, and if we do not find them helpful for describing frogs, we are quite at liberty to abandon them. We could arbitrarily divide frogs into Sex 1 and Sex 2 if we wished. However, there is one fundamental feature of the sexes which can be used to label males as males, and females as females, throughout animals and plants. This is that the sex cells or 'gametes' of males are much smaller and more numerous than the gametes of females. This is true whether we are dealing with animals or plants. One group of individuals has large sex cells, and it is convenient to use the word female for them. The other group, which it is convenient to call male, has small sex cells. The difference is especially pronounced in reptiles and in birds, where a single egg cell is big enough and nutritious enough to feed a developing baby for several weeks. Even in humans, where the egg is microscopic, it is still many times larger than the sperm. As we shall see, it is possible to interpret all the other differences between the sexes as stemming from this one basic difference.

In certain primitive organisms, for instance some fungi, maleness and femaleness do not occur, although sexual reproduction of a kind does. In the system known as isogamy the individuals are not distinguishable into two sexes. Anybody can mate with anybody else. There are not two different sorts of gametes—sperms and eggs—but all sex cells are the same, called isogametes. New individuals are formed by the fusion of two isogametes, each produced by meiotic division. If we have three isogametes, *A, B*, and *C, A* could fuse with *B* or *C, B* could fuse with *A* or *C*. The same is never true of normal sexual systems. If *A* is a sperm and it can fuse with *B* or *C*, then *B* and *C* must be eggs and *B* cannot fuse with *C*.